S403011 Machine Learning Project: Weather Prediction

Total: / 30 points

Team :

Criterion	**Excellent (4)**	**Good (3)**	**Fair (2)**	**Poor (1)**
1. Pre-processing (3 pts)	Correct cleaning; missing values, outliers, and scaling handled thoughtfully with justification.	Appropriate cleaning with minor issues; some justification.	Basic cleaning attempted; inconsistent handling of missing values or outliers.	Little or no cleaning; preprocessing errors affect results.
2. Model Training (3 pts)	Multiple appropriate models implemented correctly; hyper-parameters tuned, and reproducible.	Models trained appropriately; limited tuning or minor coding issues.	Few models tested; little attention to parameter tuning.	Incorrect or incomplete model training.
3. Model Diagnostic (3 pts)	Thorough evaluation using proper metrics; errors interpreted correctly; diagnostic plots used effectively.	Evaluation mostly correct; some interpretation missing or shallow.	Limited evaluation; metrics used but not well understood.	No real diagnostics; wrong metrics or misinterpreted results.
4. Model Comparison (3 pts)	Fair and well-argued comparison of models; clear explanation of performance differences.	Comparison of models included; some reasoning missing.	Limited or unclear comparison; weak justification.	No meaningful model comparison.
5. Model Recommendation (2 pts)	Recommendation clearly justified by metrics, interpretability, and context.	Recommendation reasonable but lacks full justification.	Weak or unclear recommendation.	No model recommendation or unjustified choice.
6. Report Structure (2 pts)	Excellent organization with clear sections and logical flow.	Good structure; minor clarity issues.	Some structure but sections confusing or incomplete.	Disorganized or hard-to-follow report.
7. Problem Understanding (2 pts)	Shows deep understanding of the task, goals, and context of weather prediction.	Shows good understanding; minor gaps in explanation.	Basic understanding with limited context.	Misunderstands problem or context.

8. Exploratory Data Analysis (3 pts)	Thorough EDA; insightful patterns identified; findings used for modeling.	Good EDA with useful plots; some insights missing.	Limited EDA; mostly descriptive with few insights.	No or poor EDA.
9. Relevant Plots (2 pts)	Plots clear, labeled, and relevant; support interpretation effectively.	Plots mostly clear; some labeling or relevance issues.	Few or unclear plots; limited interpretive value.	Missing or incorrect visualizations.
10. Language and References (2 pts)	Writing is clear, professional, and well-edited; sources correctly cited.	Mostly clear writing; minor errors; adequate referencing.	Understandable but contains errors; references incomplete.	Poorly written; no or incorrect references.
11. Python Code Structure (2 pts)	Clean, and well-documented code; easy to follow and rerun.	Readable code with minor issues.	Some organization but lacks clarity or comments.	Disorganized, hard to read, or non-functional code.
12. Choice of Libraries (1.5 pts)	Uses efficient and appropriate libraries; demonstrates solid understanding.	Libraries used correctly; minor redundancy.	Limited or inefficient use of libraries.	Inappropriate or excessive library use.
13. Creativity (1.5 pts)	Shows originality.	Some creative approaches beyond class examples.	Minor attempts at innovation.	No originality; entirely standard approach.

Total: 30 points